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Transdermal absorption of $^{13}C_4$-perfluorooctanoic acid ($^{13}C_4$-PFOA) from a sunscreen in a male volunteer – What could be the contribution of cosmetics to the internal exposure of perfluoroalkyl substances (PFAS)?

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ABSTRACT

Per- and polyfluoroalkyl substances (PFAS) are a complex group of man-made chemicals with high stability and mobility leading to ubiquitous environmental contamination and accumulation especially of some long-chain perfluoroalkyl acids (PFAA) in humans. While dietary intake is the main route of exposure, transdermal uptake from cosmetic products usually is considered negligible. However, PFAS are present in a part of these products, and recent epidemiological studies have provided evidence for relevant uptake via this route. The crucial question is whether PFAA in cosmetic products can cross the human skin barrier. A defined amount (110 µg) of $^{13}C_4$-perfluorooctanoic acid ($^{13}C_4$-PFOA) was mixed into a sunscreen (30 g) which was applied on the whole skin of a volunteer. The plasma concentrations of $^{13}C_4$-PFOA were determined in serial blood samples taken over 115 days using UHPLC-MS/MS and $^{13}C_2$-PFOA as internal standard. After application, $^{13}C_4$-PFOA plasma levels increased continuously, reaching levels of 3, 56 and 118 ng/L after 6 h, 3 days and 10 days, respectively. A maximum level of 132 ng/L was measured 22 days after application, representing 9.4 % of the PFOA level resulting from the volunteer's background exposure (1400 ng/L, equivalent to 1.4 ng/mL). In the following weeks, the levels slightly decreased with an estimated half-life of 1.8 years. The best estimate for the fraction absorbed may be 1.6 % of the dose, using a volume of distribution of 0.17 L/kg body weight. For PFOA mixed into a sunscreen, this experimental approach demonstrates a significant uptake of a PFAA via transdermal absorption in humans. In the past, some cosmetic products contained relevant PFAA levels as contaminants/impurities of PFAS added as active ingredients. Depending on these levels and the use (frequency, skin area involved), it is plausible that this route of exposure has contributed to the internal exposure to PFAA, as already suggested by epidemiological observations.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are a complex group of man-made chemicals (OECD 2021). A subgroup, the perfluoroalkyl acids (PFAA), are composed of a fluorinated carbon backbone of varying length, primarily terminated by a carboxylate (perfluoroalkyl carboxylic acids, PFCAs) or a sulfonate (perfluorosulfonic acids, PFSAs) as functional group. These compounds have amphiphilic properties with water and oil repellency, and have an extremely high chemical and thermal stability. For decades, PFAS have been used for the production of many consumer products like breathable textiles, protective coatings for paper or firefighting foams (Glüge et al. 2020). Due to their high mobility, PFAS are released from the production sites and from these everyday objects, and have caused ubiquitous contamination of the environment including the food chain (Sunderland et al. 2019; Evich et al. 2022).

Humans are primarily exposed to PFAS by consumption of food and drinking water. Due to a missing metabolic degradation pathway and low urinary excretion, some of the compounds exhibit a very high persistence in humans, with half-lives on the order of years leading to a distinct accumulation of PFAS in the body from the daily low level exposure. The resulting internal exposure especially to long-chain PFAA in individuals can be measured in serum or plasma. Four compounds, namely the PFCAs perfluorooctanoic acid (PFOA) and perfluorononanoic acid (PFNA), and the PFSAs perfluorooctane sulfonic acid (PFOS) and perfluorohexane sulfonic acid (PFHxS), typically represent more than 90 % of detectable PFAS levels of adults in

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industrialized countries (EFSA 2020). Based on biomonitoring studies from 2007/ 2008 and onwards, median concentrations of PFOS, PFOA, PFHxS, and PFNA in the adult European population were found to be 7.7, 1.9, 0.67 and 0.61 ng/ml, respectively (EFSA 2020).

In order to identify other sources contributing to the internal exposure of PFAS besides the dietary intake, the ingestion via house dust and the inhalation via indoor air was estimated by Poothong et al. (2020). The authors also estimated dermal exposure, however, considering PFAS wipeable from the skin of the hands only which may originate from PFAS-containing textiles (Peaslee et al. 2020). Dermal exposure to PFASs resulting from the use of cosmetic products has thus far been considered negligible (KEMI 2021). In this article, we address the question to what extent cosmetic products can contribute to the exposure to PFAS via transdermal absorption. In this context, three questions are crucial: (a) are PFAS indeed contained in cosmetic products?, (b) do we have evidence from epidemiological studies that cosmetic products contribute to the internal exposure to PFAS?, and (c) are PFAS absorbed through the human skin from cosmetic products containing these compounds?

Regarding the question of PFAS as ingredients of cosmetics, a first article from Japan published in 2013 reported PFCA analysis (9 compounds, C6 to C14) of 24 samples that listed fluorinated compounds (e.g. polyfluoroalkyl phosphate esters, PAP) in their international nomenclature of cosmetic ingredients (INCI) labels (Fujii et al. 2013). Twenty-one of these samples contained PFCAs (13 of 15 cosmetics, and 8 of 9 sunscreens), with concentrations of total PFCAs up to 5.9 μg/g for cosmetics and up to 19.0 μg/g for sunscreens. For PFOA, the highest concentration of 5.7 μg/g was measured in a sunscreen. Since this first article, other investigations have confirmed relevant concentrations in a part of cosmetics. For the North American market, Whitehead et al. (2021) screened 231 cosmetic products for total fluorine (TF). Of the eight categories tested, foundations, mascaras, and lip products were found to contain the highest levels of TF. In 29 products (including 20 with high TF levels), the sum of PFAS concentrations ranged from 0.022 to 10.5 μg/g (average and median: 0.26 and 1.05 μg/g, respectively). The 6:2 and 8:2 fluorotelomer alcohols (FTOH), methacrylates (FTMA), and PAP were the most frequently detected PFAS. They are precursors especially of C5-C8 PFCA, as demonstrated in biodegradation studies (Butt et al. 2014). For the Swedish market, Schultes et al. (2018) analyzed 31 cosmetic products of five categories (cream, foundation, pencil, powder and shaving foam) for TF, extractable organic fluorine (EOF) as well as 39 single PFAS. Foundations and powders contained 25 different PFAS, with the most frequently detected being PFCA (PFHpA and PFHxA) and PAP. A sum of PAP concentrations up to 470 μg/g was measured in products listing mixtures of PAP as an ingredient. These studies as well as further studies with cosmetic products from the European market (Danish EPA 2018, KEMI 2021, Pütz et al. 2022) demonstrate that especially PFCAs of various chain-lengths can be present in cosmetic products as impurities or degradation products of precursor compounds that are added as active ingredients. According to the European Commission's database on cosmetic ingredients (CosIng),[1] these substances are used intentionally in cosmetic products as emulsifiers, antistatics, stabilizers, surfactants, film formers, viscosity regulators or solvents.

Regarding the evidence from epidemiological studies, Kang et al. (2016) were the first ones reporting associations between perfluoroheptanoic acid (PFHpA) and PFOS in breastmilk (n = 214) with the use of cosmetics (foundation, powder, etc.) and basic skin care products (lotion, toner, cream, etc.), respectively. A few other studies have recently confirmed associations between PFAS and the use of cosmetic products or other personal care products (PCP) in different countries (Abdallah et al. 2019, Colles et al. 2020, Chang et al. 2021, Serrano et al. 2021). The most comprehensive study is from Norway (Thépaut et al. 2021) involving 44 males and 100 females who kept detailed diaries on their PCP use for two non-consecutive days. Multivariable linear regressions were performed between each PCP category and each PFAS level (16 compounds evaluated including two PAP), and were sex-stratified when the use of PCPs was significantly different between men and women. In the females, significantly positive associations were observed for several PFAS[2] and the use of sunscreen (6:2diPAP, PFBS, PFDS, PFTrDA), rouge/powder (6:2diPAP), and facial cleaner (PFHpS, PFHxS). For the use of mouthwash, positive associations were found for PFNA and PFOA in women and for PFOSA in men. Many PFAS showed positive associations with the use of lip-gloss and lip-balm in the females, and some of the stronger associations were observed with increased uses, such as for PFHxS, PFOA, and PFNA.

Regarding the question of PFAS absorption through the human skin, a positive proof would be the missing link between the epidemiological observations and the contents of PFAS in a part of cosmetics and other PCPs. However, such investigations dealing with the everyday use of cosmetics in humans are not available. Experimental data in rodents and *in vitro* data using diffusion cells are only scarce. *In vitro* investigations using rat and human skin replicates and a 20 % aqueous solution of ammonium perfluorooctanoate revealed only a small portion of the total amount applied to penetrate by end of the 48-h exposure period through the rat skin (1.44 ± 1.13 %) and even less through the human skin (0.048 ± 0.01 %) (Fasano et al. 2005). Franko et al. (2012) investigated the dermal penetration in mice following topical application of ^{14}C-PFOA in acetone (0.5 to 2.0 %) and observed a dose-dependent increase of ^{14}C-PFOA serum concentrations. Additional studies on the *in vitro* dermal penetration using diffusion cells with mouse and human skin were performed with ^{14}C-PFOA in acetone. After a 24-h exposure, the mean fraction of absorption was about 39 % in mouse skin and about 24 % in human full-thickness and heat-separated epidermal skin samples. Furthermore, the effect of the PFOA ionization state on dermal penetration was investigated using human epidermal membranes and aqueous solutions of ^{14}C-PFOA in water (pH 2.25) or buffer (pH 5.50), revealing median permeability coefficients of 0.055 and 0.000044 cm/h, respectively. Therefore, at the pH of the stratum corneum of about 5.5 when PFOA is almost fully ionized, its transdermal absorption from an aqueous solution was found to be more than 1000-fold lower than that of the non-ionized form with a much higher lipid solubility at pH 2.25 (Franko et al. 2012). According to the pH-partition hypothesis (Shore et al. 1957), the passive transport of charged chemical species across biological membranes is small, owing to their poor solubility in lipids.

However, the relevance of these investigations with respect to the application of a PFAS containing matrix of a cosmetic product on human skin is limited. Under these conditions, neither acetone nor water are relevant solubilizing agents, and diffusion cells are not expected to represent the *in vivo* conditions of the skin. Considering the results of epidemiological studies and content data on PFAS in cosmetic products, studying the dermal permeation potential of PFAS in humans is an urgent issue, as also addressed by Ragnarsdóttir et al. (2022) in a recent review. Therefore, we decided to perform an experimental investigation with a controlled dermal application of a sunscreen containing a defined concentration of $^{13}C_4$-PFOA in a voluntary adult.

[1] https://ec.europa.eu/growth/sectors/cosmetics/cosmetic-ingredient-database_en.

[2] Abbreviations of compounds mentioned in the following: PFBS: Perfluorobutanesulfonate; PFDS: Perfluorodecanesulfonate; PFTrDA: Perfluorotridecanoate; PFHpS: Perfluoroheptanesulfonate; PFOSA: Perfluorooctanesulfonamide.

2. Materials and methods

2.1. Concept and conduct of the pilot investigation on transdermal absorption

PFOA was chosen as model compound for this investigation for several reasons: it is one of the relevant PFAS with respect to the accumulation in humans, it is used in the *in vivo* and *in vitro* studies of Fasano et al. (2005) and Franko et al. (2012), and it was found in relevant concentrations in certain cosmetic products (see introduction). Sunscreen was chosen as the cosmetic product, as its application – for a worst-case consideration – can involve more or less the whole skin surface and is expected to be used repeatedly during summer months.

The volunteer for this self-dosing investigation was the medical head of the unit "Risks of subpopulations and human studies" (K.A., age 66 years, body weight 80 kg, body length 1.84 m, surface area about 2.0 m^2, no skin abnormalities).[3] According to our PFAS analysis, he had a current PFOA plasma level of 1400 ng/L, equivalent to 1.4 ng/mL. Assuming an equal distribution of PFOA in the body (same levels in plasma and tissues, as a worst case assumption for the body burden) results in a body burden of roughly 112 μg PFOA. As only a low proportion was expected to reach the body, a dermal dose in this order of magnitude was considered adequate for the dermal PFOA application. However, the expected increase of plasma levels after application would be too low for a reliable information in case of use of unmarked PFOA. Therefore, the compound containing four stable isotope C atoms ($^{13}C_4$-PFOA) was used as the ingredient of the sunscreen, whereas $^{13}C_2$-PFOA was used as internal standard for laboratory quantification in the plasma samples.

For the dermal application, a single dose of 110 μg $^{13}C_4$-PFOA (2.2 mL in methanol, 50 mg/L) was dried and re-dissolved in 2 mL ethanol, and then mixed into 30 g of a commercial sunscreen (for the list of ingredients see supplemental Figure S1). The resulting $^{13}C_4$-PFOA concentration of 3.7 μg/g sunscreen was comparable to the maximum of 5.7 μg/g for PFOA in a sunscreen reported by Fujii et al. (2013). After taking a shower in the morning, the sunscreen was applied on the whole body surface as usually done before a sunbath. The volunteer remained undressed for four hours in order to avoid wiping off sunscreen from skin into textile of the clothes. The next shower was 48 h after the application.

Blood samples (2.7 mL EDTA blood in tubes of Sarstedt, Nümbrecht, Germany) were taken via intravenous catheter (first hours) or via venipuncture, at least 6-hourly until 36 h, at least 12-hourly until day 5, at least every-two days until day 30, weekly until day 84, and biweekly thereafter until day 115 (last sample reported here). After centrifugation of the tubes (2500 g, 10 °C), plasma was removed and frozen at −20 °C until analysis.

2.2. Chemicals

Formic acid (reagent grade ≥ 96 %) was obtained from Sigma-Aldrich (Steinheim, Germany). Ammonium acetate (p.a.) and acetic acid (96 %, p.a.) were purchased from Merck (Darmstadt, Germany). Methanol (optigrade) was purchased from VWR (Dresden, Germany) and water (ULC-MS optigrade) was from WICOM (Heppenheim, Germany). All solvents were of analytical grade. Sterile sheep serum was from Fiebig Nährstofftechnik (Idstein, Germany). The standard reference material SRM 1957 containing organic contaminants in non-fortified human serum (freeze-dried) was from National Institute of Standards & Technology (NIST, Gaithersburg, MD, USA). Solutions of perfluoro-*n*-octanoic acid (PFOA, 1.2 mL of 50 mg/L in methanol) and the stable isotope standards perfluoro-n-[1,2,3,4-$^{13}C_4$]octanoic acid ($^{13}C_4$-PFOA, termed "MPFOA" by Wellington, 1.2 mL of 50 mg/L in methanol) and perfluoro-n-[1,2-$^{13}C_2$]octanoic acid ($^{13}C_2$-PFOA, termed "M2PFOA" by Wellington, 1.2 mL of 50 mg/L in methanol) were supplied by Wellington Laboratories (Guelph, ON, Canada).

2.3. Extraction of PFOA

During this investigation lasting from January to May 2022, single samples were analyzed from time to time to get a first impression of the time-course and to decide on the further sampling schedule. For the final analysis, 37 sampling points of time were selected,[4] extracted and analyzed in duplicate in one sample batch. The plasma was thawed at room temperature and homogenized. Aliquots of 200 μL of each sample were mixed with 20 μL of 1 μg/L $^{13}C_2$-PFOA by shaking (10 min at 800 rpm). After addition of 400 μL of water/formic acid (1:1), the samples were mixed again by shaking (10 min at 800 rpm), filled up to a volume of approx. 1.7 mL with ultrapure water and centrifuged (10 min at 18000·g, 4 °C). OASIS WAX columns (150 mg, Waters, Eschborn, Germany) were pre-conditioned by rinsing with 16 mL of 0.16 % ammonia in methanol, with 4 mL methanol and with 4 mL water. The samples were loaded and the columns were washed with 4 mL of 25 mM aqueous ammonium acetate buffer (pH 4.0) and with 4 mL methanol. After removal of residual solvents with a water jet pump, the analytes were eluted with 8 mL of 0.16 % ammonia in methanol. The collected eluates were dried in a stream of nitrogen at room temperature.

2.4. Analysis by UHPLC-MS/MS

Dried samples were reconstituted with 100 μL of methanol and 100 μL of 2 mM ammonium acetate buffer. The UHPLC-MS system consisted of a UPLC Aquity I-Class (Waters) connected to a mass spectrometer QTrap6500 (Sciex, Darmstadt, Germany) equipped with an electrospray (ESI) source operating in the negative ionization mode. An HSS T3 reversed-phase column (100 mm × 2 mm, Waters) was used for chromatographic separation. The analytes were eluted with a gradient consisting of 2 mM ammonium acetate in water/methanol (95:5) (A) and 2 mM ammonium acetate in methanol (B) at a flow rate of 300 μL/min. The gradient was as follows: 0–1 min (90 % A), 1–16 min (90–40 % A), 16–17.5 min (0 % A), 17.5–19 min (90 % A). PFOA, $^{13}C_2$-PFOA and $^{13}C_4$-PFOA were monitored using the multiple reaction monitoring (MRM) mode and three characteristic fragmentation reactions for each substance (PFOA: *m/z* 412.9 → 368.9, *m/z* 412.9 → 218.8, *m/z* 412.9 → 168.8; $^{13}C_2$-PFOA *m/z* 414.9 → 369.9, *m/z* 414.9 → 218.8, *m/z* 414.9 → 168.8; $^{13}C_4$-PFOA: *m/z* 416.9 → 371.9, *m/z* 416.9 → 218.8, *m/z* 416.9 → 168.8). The amount of $^{13}C_4$-PFOA was calculated from the peak areas of the analyte signal (*m/z* 416.9 → 371.9) and the internal standard $^{13}C_2$-PFOA (*m/z* 414.9 → 369.9). The lower signals resulting from fragmentation of the fluorinated carbon chain ($^{13}C_4$-PFOA: *m/z* 416.9 → 218.8, *m/z* 416.9 → 168.8) were used as qualifier signals. The operating parameters of the mass spectrometer were: curtain gas, 20 psi; collision-activated dissociation (CAD) gas, medium; ion source temperature, 450 °C, ion spray voltage, 5500 V; ion source gas 1, 60 psi; ion source gas 2, 50 psi. The declustering potential was − 20 V. The collision energies (CE) for the $^{13}C_4$-PFOA fragmentations *m/z* 416.9 → 371.9, *m/z* 416.9 → 218.8 and *m/z* 416.9 → 168.8 were −14 V, −24 V, and −24 V, respectively, and the cell exit potentials were −20 V, −10

[3] As a physician, K.A. does not require an approval of an ethics committee for this scientific self-experimentation performed in his own responsibility. This is a kinetic study with an investigation at an internal PFAS level (long term accumulation plus controlled $^{13}C_4$-PFOA exposure) below that of 6.9 ng/mL corresponding to the Tolerable Weekly Intake (TWI) for the PFAS sum derived by EFSA (2020). Based on current scientific PFAS knowledge, this investigation is not associated with an individual risk.

[4] A part of the samples taken within the first 36 h after application were not included in the final analysis, as a six-hour interval seems sufficient to depict the absorption process during this period.

V, and −15 V, respectively. The same voltages were applied to the fragmentations of $^{13}C_2$-PFOA. The data were recorded and evaluated with Analyst 1.7.1 Software (Sciex).

2.5. Linearity of Detection, limit of quantification and quality control

For the determination of the linear range of the quantification method sheep serum samples were spiked with $^{13}C_4$-PFOA at 11 concentrations between 0.1 ng/L and 250 ng/L. The samples were worked up and analyzed as described. The limits of detection (LOD) and of quantification (LOQ) were estimated from the natural background noise in 10 human plasma samples at the retention time of $^{13}C_4$-PFOA. The LOD for $^{13}C_4$-PFOA was defined as the mean background concentration plus three times the standard deviation and the LOQ was defined as the mean background concentration plus nine times the standard deviation (Shrivastava et al. 2011).

Two quality control samples were incorporated in the sample set in order to ensure the daily accuracy of the analyses. The first material was an in house preparation of sheep serum spiked with 1 μg/L PFOA (n = 3). The second reference material was the reconstituted serum SRM 1957 from NIST (n = 3) containing 5.0 μg/L PFOA (Rodowa et al. 2021).

3. Results

3.1. Analytical method

The analyte $^{13}C_4$-PFOA was quantified using $^{13}C_2$-PFOA as internal standard. The isotopic impurity of PFOA in plasma samples of the volunteer (~1400 ng/L) at the main trace used for the quantification (*m/z* 414.9 → 369.9) was below the LOD of $^{13}C_2$-PFOA detection (see supplemental Figure S2G). The analysis of sheep serum samples spiked with different concentrations of $^{13}C_4$-PFOA showed the linearity of detection in the range of 0.5 to 250 ng/L. The LOD and LOQ values estimated from the background integral of the quantifier trace at the precise retention time in human plasma samples for $^{13}C_4$-PFOA were 1.1 and 2.6 ng/L, respectively. Exemplary chromatograms of PFOA, $^{13}C_2$-PFOA and $^{13}C_4$-PFOA in the samples taken before, and 12 h as well as 10 days after the dermal application are shown in supplemental Figure S2.

3.2. Transdermal absorption kinetics

Levels of $^{13}C_4$-PFOA and PFOA measured in plasma following the dermal application of $^{13}C_4$-PFOA in a sunscreen are displayed in Fig. 1 and compiled in supplemental Table S1. Whereas no $^{13}C_4$-PFOA was detectable before the application, a level of 3.0 ng/L was already quantifiable after six hours. During the first three days, a more or less linear increase was observed, with a daily increase of about 19 ng/L. In the following days, a continuous flattening of the increase occurred, and a level of 118 ng/L was reached after 10 days. In the following time, the scattering of the data does not allow to describe the time-course exactly. A maximum level of 132 ng/L was measured 22 days after application. For the following weeks, a tendency towards decreasing levels was observed (Fig. 1) until the end of the observation period reported in this article (115 days).

For the levels of PFOA (Fig. 1, supplemental Table S1), a mean of 1400 ± 80 ng/L (n = 37) was observed for the total period. Displaying the data in a semilogarithmic plot (supplemental Figure S3A) revealed a slight decrease with an overall half-life of about 2.6 years. Due to the scattering of the data (analytical and biological variability), a corresponding analysis of the $^{13}C_4$-PFOA data does not yield reliable results, as they very much depend on the data points included in the calculations. However, the scattering pattern of $^{13}C_4$-PFOA and PFOA was very similar after the peak phase (see Fig. 1), with a high correlation of $^{13}C_4$-PFOA and PFOA for the 9 sampling points in time from day 45 to day 115 (R^2 = 98 %). Assuming the true PFOA plasma levels to lie on a straight line after logarithmic transformation (representing a calculated

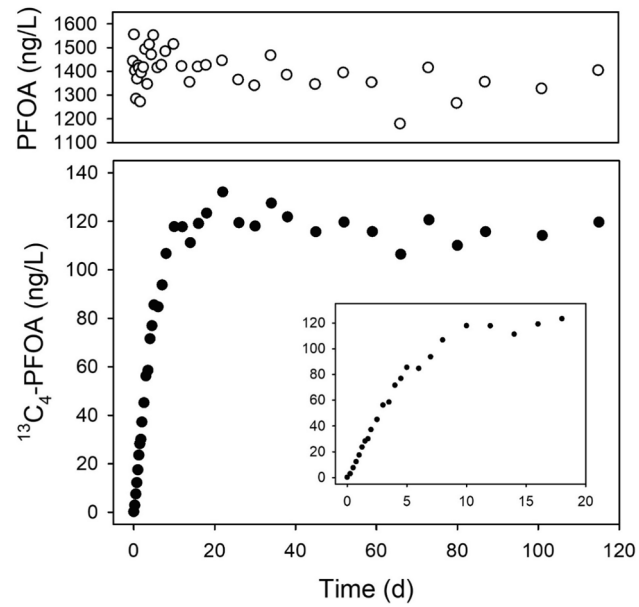

Fig. 1. Time course of the plasma levels of $^{13}C_4$-PFOA (black dots) after dermal application of $^{13}C_4$-PFOA in a sunscreen (Insert: enlargement of the time course during the first 20 days after application). White dots are the PFOA levels resulting from the background exposure measured simultaneously.

half-life of 2.6 years) allows the calculation of correction factors for each point in time (supplemental Table S1). Applying these factors to the plasma levels of $^{13}C_4$-PFOA leads to a considerable reduction of the scattering (supplemental Figure S3B). In the semilogarithmic plot for the time course, R^2 increases from 13 to 79 % if the 13 sampling points in time from day 26 to day 115 are included. This allows the calculation of a terminal half-life of about 1.8 years (elimination rate constant k_{elim} = 0.376/yr = 0.00103/d, supplemental Figure S3C) with a theoretical concentration ($c_{t=0}$) of 125 ng/L at the time of application (t = 0). Considering a simple one-compartment model with first order kinetics for absorption and elimination with the formula for the time-depending concentration $c = c_{t=0}*(\exp(-k_{elim}*t) - \exp(-k_{abs}*t))$ reveals a value for the absorption rate constant k_{abs} = 0.212/d,[5] i.e. a time of 3.3 days to absorb half of the total dose finally reaching the circulation. The resulting curve fits the values measured almost perfectly (see supplemental Figure S4 and Table S1) and has a maximum of 121.2 ng/L on day 25 after application.

Regarding the fraction absorbed of the dermally applied amount of 110 μg $^{13}C_4$-PFOA, only rough approximations can be made. Using PFOA blood distribution data (median plasma 1.60 μg/L, whole blood 0.93 μg/L; Poothong et al. 2017), the maximum plasma $^{13}C_4$-PFOA level of 132 ng/L corresponds to a level of 77 ng/L in whole blood. Considering a total blood volume of 5.46 L in the volunteer (according to the Nadler equation; Sharma and Sharma 2021), a total amount of 419 ng $^{13}C_4$-PFOA would be maximally contained in blood. Assuming a distribution within the blood only (no distribution to organs and tissues), 0.38 % of the dermally applied dose would have reached the body. Assuming an equal distribution in the body (weight 80 kg) with the concentration in whole blood, 6160 ng $^{13}C_4$-PFOA (5.6 % of the dose) would have been dermally absorbed. The best estimate for the fraction absorbed may result from using a volume of distribution of 0.17 L/kg body weight (estimated by Thompson et al. 2010, based on serum levels),

[5] Calculated by the "method of residuals", using the values measured within the first 7 days.

corresponding to 1800 ng $^{13}C_4$-PFOA (1.6 % of the dose) having reached the body.[6]

4. Discussion

4.1. Study observations

This is the first report on transdermal absorption of a PFAA (PFOA) mixed into a cosmetic product (sunscreen), using a relevant application of everyday life in an experimental approach. It reveals a significant absorption with a bioavailability of at least 0.38 % of the $^{13}C_4$-PFOA dose if a sole distribution in blood only is assumed. This is a low value if compared to the high oral PFAS bioavailability assumed in general (EFSA 2020). However, in case of high concentrations in the μg/g range in a sunscreen, a daily application during summer months would result in very high plasma concentrations. This is obvious, as already the single application of a sunscreen with a PFOA content of 3.7 μg/g leads to a maximum $^{13}C_4$-PFOA plasma level (132 ng/L) which represents 9.4 % of the long-term PFOA level (1400 ng/L) resulting from all sources of PFOA exposure in this volunteer.

With the methods used for this investigation, an exact quantification of the dermal bioavailability of $^{13}C_4$-PFOA in a sunscreen is not possible. It is very likely that a part of the absorbed $^{13}C_4$-PFOA is distributed from blood to other fluids and tissues. However, only very few reports dealing with PFAS levels in organs and blood are available for humans (Olsen et al. 2003, Maestri et al. 2006, Yeung et al. 2013), and these data suggest that the accumulation in most tissues does not exceed that in blood. Therefore, the estimation for the dermal bioavailability assuming an equal distribution in the body may be interpreted as worst-case consideration, and the result for the fraction absorbed of about 1.6 % (using an estimated volume of distribution of 0.17 L/kg body weight, Thompson et al. 2010) may be the best estimate.

In view of the time-course of $^{13}C_4$-PFOA levels observed (Fig. 1), two points are remarkable with respect to the kinetics: (a) the very slow transdermal absorption with a time window to reach the peak of about three weeks after the single application, and (b) the lack of a distribution phase with a temporary faster decrease of $^{13}C_4$-PFOA levels after reaching the maximum levels. The latter may be due to a much faster distribution within the body in comparison to the slow dermal absorption. Compared to the overall half-life of about 2.6 years observed for PFOA from background exposure, a shorter half-life of about 1.8 years was observed for $^{13}C_4$-PFOA after reaching the maximum levels. This difference is unlikely to result from an ongoing distribution process in case of $^{13}C_4$-PFOA, but likely reflects the continuous daily low level exposure to PFOA during the sampling period especially from nutrition.

Furthermore, a third point is remarkable with respect to the time-course of the $^{13}C_4$-PFOA plasma levels and its nearly perfect first order kinetics (supplemental Figure S4): The overall absorption process looks like a simple, gradient-driven diffusion through a barrier without a distinct time delay,[7] with a maximal daily uptake already reached on day 2. This makes a relatively fast uptake of the $^{13}C_4$-PFOA by the stratum corneum likely, and the shower 48 h after the administration has no visible influence on the kinetics.

On the other hand, these observations seem somewhat at odds with the relatively long time of 3.3 days to absorb half of the total dose finally reaching the circulation. The mechanisms involved in the unusually slow absorption process are currently unknown. With respect to the *in-vitro* investigations with focus on the ionization state of PFOA (Franko et al. 2012, see introduction), an absorption from an aqueous PFOA solution at the pH of the stratum corneum of about 5.5 may be considered negligible, as PFOA is almost fully ionized under these conditions. However, the ionization state is always an equilibrium between dissociated and undissociated molecules. Under the condition of the application of a sunscreen on the skin, undissociated PFOA molecules may be able to slowly cross the cellular layers of the epidermis, leading to quantifiable $^{13}C_4$-PFOA plasma levels "already" six hours after the application. These molecules disappearing from the sunscreen (shortly after application) and from the upper layers of the epidermis are replaced by molecules changing from the dissociated to the undissociated state. Considering this dynamic equilibrium, a relevant part of the $^{13}C_4$-PFOA dermally applied with the sunscreen may be able to slowly penetrate the epidermis layer by layer via passive diffusion and to reach the circulation. After about three weeks, this process obviously comes to an end. This relatively long time is expected to be also caused by the epidermal cell turnover, leading to a contraflow of layers (complete cell turnover occurs every 28 to 30 days in young adults, while the same process takes 45 to 50 days in elderly adults[8]). The part of the dose not absorbed is expected to be wiped of (by clothes and other contacts) and finally to be removed with the top outer layers of the stratum corneum during desquamation.

This experimental investigation has the character of a proof-of-principle test in a single person, using PFOA as model compound and sunscreen as model cosmetic product. It unambiguously demonstrates that relevant amounts of PFOA penetrate the skin under these conditions. Of course, a higher number of participants would provide insights to the individual variability of the absorption process, but would provide results for one compound (PFOA) in one cosmetic product (sunscreen) only. Therefore, the complex issue of PFAS as constituents in cosmetic products cannot be reasonably elucidated by performing studies in humans using certain PFAS in certain cosmetic products.

4.2. Conclusions on PFAS in cosmetic products

The important novel message is that transdermal absorption of PFOA (and likely other PFAS) from cosmetic products is not negligible *per se* and able to contribute to the overall exposure to PFAS in case of relevant levels in the cosmetic product. Considering reports on the PFAS levels in these products from recent years, a part of the products investigated contained a wide range of PFAS, e.g. FTOH, FTMA, PAP or polytetrafluoroethylene (PTFE) (Fujii et al. 2013; Danish EPA 2018; Schultes et al. 2018; Whitehead et al. 2021, see introduction). A recent analysis of the market share of PFAS-containing products in Sweden (KEMI 2021, Pütz et al. 2022) revealed that most occurred in the product category "Decorative cosmetics" (3.7 %), followed by "Skin care" (0.78 %), "Hair care" (0.65 %) and "Toiletries" (0.27 % of the products). The PFAS mentioned in turn may contain relevant amounts especially of PFCA as contaminants/impurities, with maximum levels reported in the μg/g range (e.g. Fujii et al. 2013). Altogether, the investigations on PFAS in cosmetic products reveal a very complex issue, and the relevance for the exposure to consumers is expected to depend on many factors: spectrum of PFAS and especially their PFCA contaminants/impurities used for the production in a certain country at a certain time, as well as frequency of use and size of skin area involved for a certain product.

Having this and the result of our investigation in mind, it appears plausible that the transdermal route is relevant and can contribute significantly to the internal exposure to PFAS on the population level, as already suggested from the results of epidemiological studies in different countries revealing associations between internal exposure to PFAA and the use of certain cosmetic products (Kang et al. 2016, Abdallah et al. 2019, Colles et al. 2020, Chang et al. 2021, Serrano et al. 2021, Thépaut et al. 2021). This awareness applies at least to the past. Currently,

[6] The amount of ^{13}C-PFOA in the body is the product of the volume of distribution (13.6 L in case of the volunteer) and the maximum ^{13}C-PFOA level in blood plasma (132 ng/L).

[7] This process is likely to include a simultaneous process of distribution in the body.

[8] Layers of the skin: https://training.seer.cancer.gov/melanoma/anatomy/layers.html.

cosmetic industry in part is working on a phase out of PFAS in cosmetics (KEMI 2021). Furthermore, e.g. the European Union (EU) has established maximum levels of 25 ppb (μg/kg) for PFOA and its salts[9] in substances, mixtures and articles since July 2020 as well as for the sum of C9-C14 PFAA and its salts[10] in substances, mixtures and articles from February 2023 on. Therefore, the transdermal uptake of long-chain PFAA resulting from the use of cosmetics containing these compounds as contaminants/impurities is expected to be reduced currently and in the future.

However, a yet unanswered question remains, namely the role of the several PFAS like FTOH or PAP added as active ingredients to cosmetic products which therefore may contain the compounds in high concentrations, e.g. up to 470 μg/g for the sum of PAP (Schultes et al. 2018). They in part have also been measured in relevant concentrations in humans (Thépaut et al. 2021), and transdermal absorption is not unlikely. The compounds are not only possibly contaminated with PFAA, but are also possible precursors of PFAA in case of formation as spontaneous degradation products or by metabolizing enzymes in skin or organs (Vestergren et al. 2008, D'eon and Mabury 2011, Butt et al. 2014, KEMI 2021, Evich et al. 2022).

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Klaus Abraham: Conceptualization, Investigation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Bernhard H. Monien:** Methodology, Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data included in the Supplemental Material

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2022.107549.

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